

Food Safety and FDM 3D-Printed Parts: A Risk-Relative Assessment

TL;DR

- The "inherently unsafe" narrative is **not supported by measured data**: peer-reviewed work shows lead migration from brass-nozzle prints in the single-digit-nanogram range per part (hundreds of times below FDA's daily lead limits), and controlled lab work shows PLA and PETG prints can be cleaned to safe microbiological levels with warm soapy water — while accepted kitchen items such as sponges and used plastic cutting boards routinely carry far higher bacterial loads.
- The one legitimate, non-equivalent residual concern is **unknown additives/colorants in non-certified filaments plus layer-line and internal-void geometry on poorly designed or unsealed prints** — but these are design and material-selection problems, not an intrinsic property of FDM.
- Net assessment: a properly designed, well-printed, and routinely cleaned FDM part in a food-grade polymer poses risk **comparable to or below** everyday accepted items (wooden cutting boards, sponges, reusable bottle seals, restaurant menus); there are **no documented foodborne-illness outbreaks** attributed to 3D-printed kitchen items in the surveillance literature.

Key Findings

1. **Bacteria and layer lines**: Layer-line valleys do preferentially accumulate biofilm, but they are large channels (~tens of microns), not microscopic pores, and they are cleanable. The most-cited concern study (Hall et al., *Frontiers in Microbiology*, 2021) found biofilm thickest between layers — but also found wide variation by polymer and surface roughness. A multi-month controlled lab study (Neamah et al., IEEE) found PLA/PLA+ and PETG can be cleaned to safe levels with ~120 °F water and dish soap.
2. **Lead from brass nozzles**: The best peer-reviewed measurement (Rindelaub et al., *PLOS ONE*, 2019) found lead as the only detected heavy metal, at 4.95–7.37 ng per 32.5 cm² device for brass nozzles vs. 0.60–0.69 ng for steel — 7–12× higher with brass, but still single-digit nanograms. An independent engineering estimate put transfer at ~200 ng per part. Both are far below FDA's Interim Reference Level for lead.
3. **Plastic degradation is not unique to 3D prints**: Scratched non-stick (PTFE) pans shed thousands to millions of microplastic particles; used/scarred plastic cutting boards harbor and grow bacteria. The same "if water gets in, so does soap" logic and the same degradation risks apply across the kitchen.
4. **Comparative bacterial landscape**: Kitchen sponges carry up to $\sim 5.4 \times 10^{10}$ cells/cm³; used scarred plastic cutting boards retain and even grow pathogens overnight; restaurant menus

and touchscreens are measurable vectors. A cleaned FDM part is not an outlier in this landscape.

5. **FDA stance is precautionary, not measurement-based, and is applied unevenly:** FDA does not "approve" or certify any finished food-contact article — 3D-printed or injection-molded. It regulates the substance, not the object. No finished wooden cutting board or Tupperware tub is "FDA-approved" either.

Details

1. Layer lines and bacterial retention

The claim: Layer-line grooves trap bacteria that cannot be cleaned out.

What the studies actually show. The most frequently cited source is Hall, Palmer, Ji, Ehrlich & Król, "Bacterial Biofilm Growth on 3D-Printed Materials," *Frontiers in Microbiology* 12:646303 (2021), from Drexel University. Using *E. coli*, *P. aeruginosa*, and *S. aureus* on eight commercial PLA polymers (MakerBot Replicator2, 0.2 mm layers, 15% infill), they reported that biofilms were thickest "in between the layers of the 3D printed materials," and that biofilm formation "depends on surface structure, hydrophobicity, and that there was a wide range of antimicrobial properties among the tested polymers." Critically, this is a study about *relative biofilm attachment on uncleaned, unsealed surfaces* — it does not demonstrate that cleaned prints remain hazardous, and it found several PLA formulations had antifouling characteristics. The often-quoted line that biofilm "grows thickest in the valleys between printed layers" is real, but it describes growth conditions, not a cleaning failure. (nih)

The geometry matters. Independent lab/engineering analysis (a University of Utah Valley microbiology/engineering writeup) using scanning electron microscopy reported that FDM prints have "layer lines, cusps, and dimples, but not pores," with layer lines "10 microns in width and about 10 microns deep" — enormous relative to a ~1-2 micron bacterium and, more importantly, relative to a surfactant molecule. The "cleaning admits water" logic is sound: if a crevice is wide enough for a bacterium and moisture to enter, it is wide enough for soap solution to enter and lift it out. Bacteria also require moisture to persist, so a dried, clean part is not a growth substrate.

The strongest direct evidence for cleanability is the work by H. Neamah and colleagues, "Study on the Sanitization Efficacy for Safe Use of 3D-Printed Parts for Food and Medical Applications" and its follow-ons (IEEE), which built on a 4-month lab study and added 3 more months in household kitchens. Their conclusion: "3D printed parts of PLA/PLA+ (Polylactic Acid), and PETG (Polyethylene terephthalate glycol) can be cleaned to safe levels using warm water (120 °F), and non-concentrated dish soap," verified via Petri-dish culture and protein-residue testing.

ResearchGate

Honest caveat: Internal voids/low-infill interiors and chemically unsmoothed surfaces are genuinely harder to clean. The remedies (100% infill or wall-only solid geometry on food-contact

zones, low layer height, chemical smoothing, or a food-safe coating) are well established and effective.

2. Lead contamination from brass nozzles

The claim: Brass nozzles leach lead into food-contact surfaces.

The measured reality. Standard nozzles use C36000 free-machining brass, which contains roughly 2.5–3.7% lead by weight. The key peer-reviewed measurement is Rindelaub, Baird, Lindner & Strantz, "Identifying extractable profiles from 3D printed medical devices," *PLOS ONE* 14(5):e0217137 (2019). Printing 32.5 cm² coupons and performing ICP/MS on water extracts, they found, verbatim: "Of the 15 elements targeted by ICP/MS analysis, only lead (Pb) was observed in the printed device extracts. Pb was not observed above the method detection limit (0.35 ng/device) in the filament extracts." Brass-nozzle devices measured ~0.98–1.46 ng/g, or 4.95–7.37 ng per device; steel-nozzle devices measured 0.11–0.13 ng/g, or 0.60–0.69 ng per device. The authors state: "The devices printed using the brass nozzle had 7–12 times greater concentrations of Pb compared to those printed using a steel nozzle." SEM-EDS showed only ~2% of extracted particles contained nozzle-derived metal; the large majority were carbon (polymer). The same study found a strong mitigation: "The application of an annealing procedure to printed devices led to a substantial decrease in extractable components by as much as a factor of 50." (plos)

Putting the number in context. The worst-case figure is ~7.4 ng (0.0074 µg) of total extractable lead per 32.5 cm² part. Per Flannery & Middleton, *Regulatory Toxicology and Pharmacology* (Epub 2022 Jun 8; PMID 35690180), "the FDA is updating its interim reference levels (IRLs) for lead from food to 2.2 µg/day for children and 8.8 µg/day for females of childbearing age" — down from the prior 3 and 12.5 µg/day, aligning with CDC's 2021 blood-lead reference value of 3.5 µg/dL. Even if the *entire* extractable lead content of a part transferred into a single day's food, ~7.4 ng would be roughly 0.3% of the children's 2.2 µg/day IRL. An independent engineering estimate (Marzocchi, 2023) using deliberately pessimistic assumptions arrived at ~200 ng (0.2 µg) of lead per part — still under 10% of the children's IRL and well under the adult figure. The arXiv mass-spectrometry radiopurity study of FDM parts (for rare-event physics detectors) similarly treats nozzle-derived metal as a trace-level, characterizable quantity rather than a bulk contaminant. (Marzocchi)

Honest caveat: The Rindelaub study confirms brass nozzles *do measurably increase* lead — so the strongest version of the pushback ("zero detectable lead") is incorrect. The accurate framing is that lead is detectable but at nanogram-per-part levels far below any toxicological threshold. Abrasive filaments (metal-filled, carbon-fiber, some glitter/glow formulations) accelerate nozzle wear and are a separate, real concern; for food contact, a stainless-steel nozzle plus non-abrasive food-grade filament removes even this trace.

3. Plastic degradation over time

Degradation is a general property of kitchen plastics, not a special 3D-printing risk. The University of Newcastle study (Luo et al., *Science of the Total Environment*, 2022, DOI

10.1016/j.scitotenv.2022.158293) used Raman imaging and reported: "Broken coating may lead to release of 2,300,000 microplastics and nanoplastics. A surface crack could leave behind approximately 9100 plastic particles" — from roughly 30 seconds of dry-mixing on a PTFE non-stick pan. A Plymouth Marine Laboratory study (Cole et al.) found plastic cutting boards, utensils, and containers all shed microplastics, and that PTFE cutting boards can contribute up to 68.9 microplastics per gram of meat. The "cleaning admits water/soap" argument applies equally to scratched plastic and scarred wood. There is no evidence that degraded FDM PLA/PETG behaves worse than these accepted, ubiquitous items; the degradation discourse around 3D printing is disproportionate to the data.

4. Comparative risk landscape

- **Kitchen sponges** are repeatedly the most contaminated kitchen item. Cardinale et al., *Scientific Reports* 7:5791 (2017), found "local densities of up to 5.4×10^{10} cells per cm³, and confirmed the dominance of Gammaproteobacteria," with culture-based counts of $\sim 10^7$ – 10^9 CFU per sponge — and found that routine sanitizing did not reliably reduce, and sometimes enriched, risk-group-2 species. Field studies of food establishments report sponge counts of 7.4–12.4 log CFU/mm³.
- **Cutting boards:** The classic Cliver/Ak/Kaspar work (University of Wisconsin, *Journal of Food Protection* 57:16–22 and 57:23–30, 1994) found ~99.9% of bacteria died on wooden boards within 3 minutes, while bacteria survived and even multiplied overnight on plastic; knife-scarred plastic boards were "impossible to clean and disinfect manually" once food residue was present. A 2025 *Journal of Food Protection* study reaffirmed maple's bactericidal effect versus HDPE. This is directly relevant: the everyday, regulator-recommended plastic cutting board is a worse bacterial reservoir than wood once scarred — yet it is universally accepted.
- **Restaurant menus / touchscreens:** Alsallaiy et al. (*Journal of Food Safety*, 2016) found menus averaged ~28 CFU/15 cm² during busy periods, with bacteria transferring to hands and surviving 24–48 h. Fast-food self-order touchscreens carried 0.07–5.7 CFU/cm² (BMC Research Notes). These uncleaned vectors are tolerated daily.
- **Documented illness:** A search of the foodborne-outbreak literature surfaces **no documented foodborne-illness outbreak attributed to a 3D-printed kitchen utensil or food-contact part**. Per CDC's MMWR Surveillance Summaries Vol. 74/No. 1 (Mar 13, 2025): "Approximately 800 foodborne illness outbreaks occur in the United States each year. These outbreaks include approximately 15,000 illnesses, 800 hospitalizations, and 20 deaths." Reported contributing factors are overwhelmingly hand-hygiene failures, sick food workers, temperature abuse, and cross-contamination — not utensil material.

5. Regulatory / FDA stance

FDA does **not certify or approve any finished food-contact article**, whether 3D-printed, injection-molded, wooden, or ceramic. It regulates *food-contact substances*: a substance must be

listed in 21 CFR (e.g., PET under 177.1630), be Generally Recognized as Safe (GRAS), fall under a Threshold of Regulation exemption, or be cleared via a Food Contact Notification. PLA and PET/PETG base resins are FDA-compliant for food contact; that compliance attaches to the resin, not the printed object. Industry and university-extension sources (Polymaker; Oklahoma State University Extension fact sheet) correctly note that "no 3D printing material holds FDA food-safe certification" — but the same is literally true of a wooden cutting board or a Tupperware tub: there is no FDA certificate for those finished objects either. The "FDA hasn't approved it" framing is therefore precautionary and applies a standard (finished-article certification) that **does not exist for any kitchen item**. The EU is more prescriptive: Regulation (EU) 10/2011 sets an Overall Migration Limit of 10 mg/dm² ($\approx$ 60 mg/kg food) plus specific migration limits, including a specific element limit for lead — but again applies to a material/article test, not a 3D-printing-specific ban.

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Recommendations

Stage 1 — Material and hardware selection (highest value, do this first):

- Use a base polymer that is FDA-compliant/EU-listed for food contact: PLA, PETG, or PP, in a **natural/uncolored, additive-disclosed food-grade grade**. Avoid unknown colorants and recycled/blended filaments — this is the single most important step and the only genuinely non-equivalent residual risk versus conventional kitchenware.
- Use a **stainless-steel nozzle** for any food-contact print. This removes even the nanogram-level brass-lead contribution for $\sim$ \$10.

Stage 2 — Print and post-process:

- Solid geometry on food-contact surfaces (100% infill or wall-only thick shells; no exposed low-infill voids), low layer height ($\leq$ 0.1–0.15 mm), and high wall overlap to minimize inter-layer channels.
- For repeated, wet, or long-contact use, apply a food-safe epoxy/polyurethane coating or chemically smooth the surface — this converts it to effectively non-porous. Annealing additionally cuts extractables (up to $\sim$ 50 $\times$ per Rindelaub et al.), though it can warp dimensionally accurate parts.

Stage 3 — Use and clean:

- Hand-wash with warm ($\sim$ 120 °F) water and dish soap after each use. Do not put PLA in a dishwasher (it deforms near 55–60 °C); PETG and PP tolerate more heat.
- Prefer uncoated PLA for **dry, brief, or cool contact** (cookie cutters, dry-goods scoops, single-use molds) over hot/wet/acidic long-contact uses.

Benchmarks that would change this advice:

- Print is in an uncertified/colored filament with undisclosed additives → treat as **not food-safe** (additive/colorant migration is the real unknown).
- Part has open internal voids or is visibly degraded/cracked and uncoated → retire it, exactly as you would a deeply scarred plastic board or a flaking non-stick pan.
- Hot (>60 °C for PLA), acidic, or fatty long-duration contact is required → switch material (PP/PETG) or use a coating/different process.

Caveats

- **Source-quality note:** Many "layer lines harbor bacteria / cannot be cleaned" claims online trace to vendor blogs and a single relative-attachment study; they are frequently overstated, and the controlled sanitization data contradict the "cannot be cleaned" version. Conversely, some pushback sources overstate "zero lead" — the peer-reviewed data show detectable-but-trivial lead.
- The Rindelaub lead study was an *extractable* (aggressive-solvent) study on medical-device coupons, not a food-simulant migration study; true food-contact migration is likely lower, but the study also carries a commercial-lab competing-interest disclosure (two authors employed by an E&L testing lab). The Neamah sanitization work is IEEE conference-level rather than a high-impact journal and would benefit from independent replication.
- "No documented outbreaks" reflects absence of evidence in surveillance systems, partly because home-printed utensils are not separately tracked; it is strong but not absolute proof of safety.
- This assessment covers **FDM/FFF thermoplastic** prints only. **SLA/resin prints are a different and genuinely higher-risk category** — most photopolymer resins are not food-safe in cured or uncured form — and should not be used for food contact without specifically certified materials.

Bottom line: The evidence does not support treating properly made and cleaned FDM food-contact parts as "inherently unsafe." Measured lead transfer is in the nanogram range (orders of magnitude below regulatory limits), bacterial retention is cleanable to safe levels, degradation is no worse than accepted plastics, and there are zero documented illness cases — placing a well-executed FDM part at or below the everyday risk of sponges, scarred plastic boards, and other tolerated kitchen items. The real, non-equivalent risks are narrow and controllable: uncertified colored/additive-laden filament, brass nozzles (trivially swapped for steel), and unsealed internal voids on poorly designed prints.